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Cells image generation method based on VAE-SGAN

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Abstract

Although it has entered the era of big data, the image data with important research value in the field of medical imaging is often scarce due to various reasons such as difficult acquisition. To increase the image data in these fields, people have developed computer simulations to generate images Methodological research. At present, there are two main types of image generation models with potential, namely, Variational Autoencoders (VAE) [1] and Generative Adversarial Networks (GAN) [2]. In the process of classifying white blood cells, there are often insufficient sample data and unbalanced data. How to effectively expand the sample size and solve the imbalance of training data has always been a hot issue in research. In order to solve this problem, this paper proposes an image generation method based on the fusion network of VAE and GANs, which avoids the shortcomings of the two generation methods when used alone, and makes the generated images have higher generation quality.

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Keywords: Variational Autoencoders, GAN, white blood cells, image generation.

1. Introduction

In clinical medicine, blood is significantly important to human. Human blood includes the cellular and liquid parts. There are red blood cells, white blood cells and platelets in the cell part [3]. These three types of cells are called blood cells. All three of them have different physiological functions in the human body, and together support the healthy growth of life. The main function of the whole white blood cells system is to protect the body from the invasion of foreign substances [4]. They are responsible for identifying and phagocytosing abnormal cells and acting as destroyers of viruses and bacteria. Generally, the number of white blood cells, one of the main indicators of routine blood tests, is maintained at a certain level, and it is often used as an important basis for the diagnosis of

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systemic diseases [5]. The morphology and size of white blood cells vary, and generally include lymphocytes, neutrophils, basophils, eosinophils and monocytes [6]. They have their own physiological functions, but they also cooperate with each other to protect the body from viruses or bacteria.

More influential in the generative models are VAE, PixelCNN [7], Glow [8], and GAN. Among them, GAN is composed of two neural networks, one is called generator and the other is called discriminator. The purpose of the generator is to generate samples that can deceive the discriminator [9]. What the discriminator needs to do is to distinguish the real samples from the generated samples. This kind of confrontation game makes the generator and discriminator continue to improve performance. After reaching the Nash balance, the generator can output images similar to real images. However, this Nash balance only exists in theory, and the actual GAN training is accompanied by some problems. One is GAN training instability and the other is mode collapse. As one of the most popular deep generation models in recent years, the Variational Autoencoder has made significant improvements in completing the representation learning of high-dimensional information and the completion of data generation at the same time, which has greatly promoted the research of deep generation models. However, because it uses the mean square error as a loss function, the resulting image lacks details and the picture is blurred. In this paper, the joint training of VAE and GANs can stably generate real cell pictures.

2. Related Work

2.1. VAE

Different from GAN, VAE adds some restrictions in the encoding process, which makes the implicit vectors generated by the network follow the standard normal distribution [10]. At this time, instead of generating an implicit vector each time, two vectors are generated, one for the mean and one for the standard deviation, and then the two vectors are used to synthesize the implicit vector, that is, a standard normal distribution is used first. Multiply the standard deviation and add the mean. The network structure of VAE is shown in Figure 1. VAE can solve the problem of continuous data generation [11]. In terms of inference problems, VAE has the advantages of fast training, high stability, high diversity, and high image reconstruction quality, but the generated picture has low clarity [12].

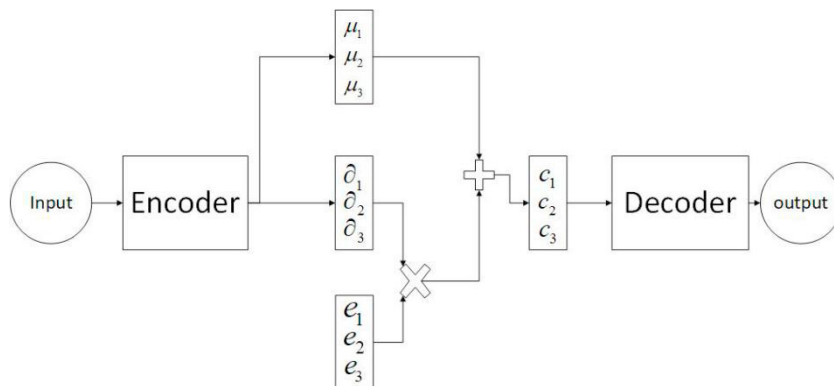


Fig. 1. VAE structure schematic diagram

VAE adds a priori $p(z)$ to the posterior distribution and optimizes it through KL divergence. This requires that $p(z)$ must have an explicit probability distribution. The objective function of VAE is:

$$\begin{aligned}
& E_{x \sim p_d(x)}[-\log p(x)] < E_x[E_{q(z|x)}[-\log(p(x|z))] + E_x[KL(q(z|x) || p(z))]] \\
& = E_x[E_{q(z|x)}[-\log p(x|z)]] - E_x[H(q(z|x))] + E_{q \in}[-\log p(z)] \\
& = E_x \left[E_{q(z|x)}[-\log p(x|z)] - E_x \left[\sum_i \log \sigma_i(x) \right] \right] + E_{q \in}[-\log p(z)] + const \\
& = Reconstruction - Entropy + CrossEntropy(q(z), p(z))
\end{aligned} \tag{1}$$

2.2. StyleGAN

Through ProGAN [13] network embedding technology, even the low-resolution image can learn its basic features to create the basic part of the image, and as the resolution increases and the number of training increases, more and more details can be learned. However, ProGAN directly generates pictures step by step, it is impossible to know what features it has learned at each level. This results in the very limited ability of ProGAN to control the generation of specific features of an image. In other words, these characteristics are interrelated, so trying to adjust the input, even a little, usually affects multiple characteristics at the same time. In order to better control the output picture, this article uses StyleGAN [14].

In the generator network architecture, a style line graph and 100-dimensional vector noise are taken as inputs. They first enter the convolutional layer and the deconvolution layer, and then merge the generated feature maps. 7 layers of convolution and deconvolution are further performed on these function diagrams, and the final output is an image. For the distinguisher, we use a structure similar to StyleGAN, as shown in Table 1 [15]. A large number of structural lines and images are input into the network. The specific architecture of the generator in StyleGAN is shown in Figure 2.

Table 1. Style GAN discriminates network parameter setting table

Style-GAN(D)	conv	conv	conv	conv	conv	fc
Input Size	128	64	32	16	8	—
Kernel Number	64	128	256	512	128	1
Kernel Size	5	5	3	3	3	—
Stride	2	2	2	2	1	—

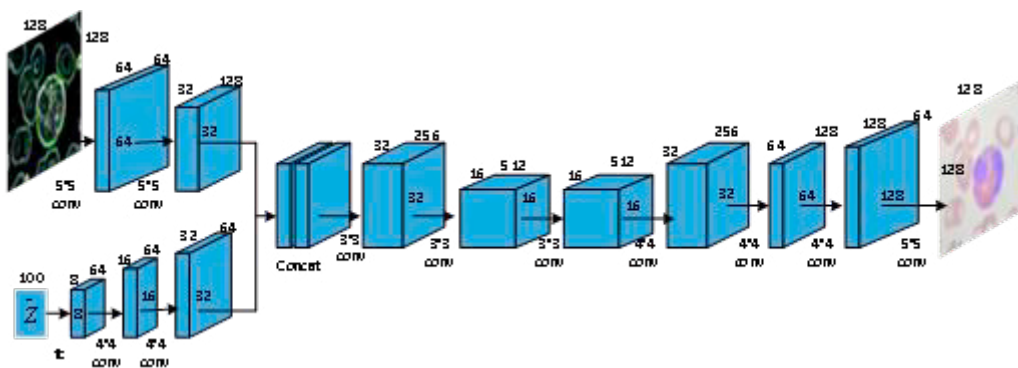


Fig. 2. Generator specific architecture diagram in StyleGAN

2.3. VAE and StyleGAN

We combine VAE and StyleGAN into an unsupervised generative model, which generates and compares data set samples while learning coding. Figure 3 is the complete architecture of the joint model. After coding by encoder, a

latent vector z containing features is generated, where z follows a Gaussian distribution, and z generates image data x' through a generator (decoder). Because the loss function combines the two parts of the decoder and the generator, there are actually two parts of x' here. One part is obtained from the decoder through the Gaussian distribution vector, and the other is generated by the generator according to z . Finally, the generated sample x' and the original data x are sent to the discriminator to judge the authenticity, so as to achieve the combination of VAE and GAN.

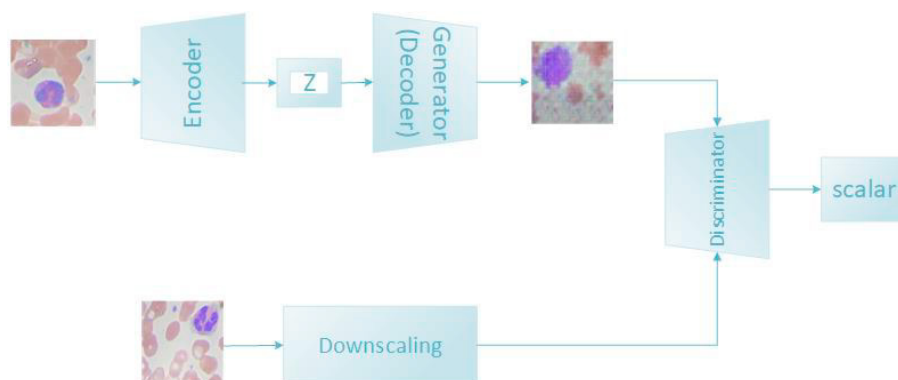


Fig. 3. The complete architecture of the joint model

We use the encoder network to map the real image to a latent vector z , where z follows a Gaussian distribution, and the characteristics of the image match the given latent vector. As shown in Figure 4, contrary to feeding the random vector directly into the generator, the image is decoded and input, and then projected onto the intermediate latent space w through the mapping network. Real samples can be discontinuous (normal GANs cannot be trained and optimized), but z is continuous. We can fine-tune z to generate more realistic samples. In other words, the new model avoids the problem that GAN cannot generate discrete samples.

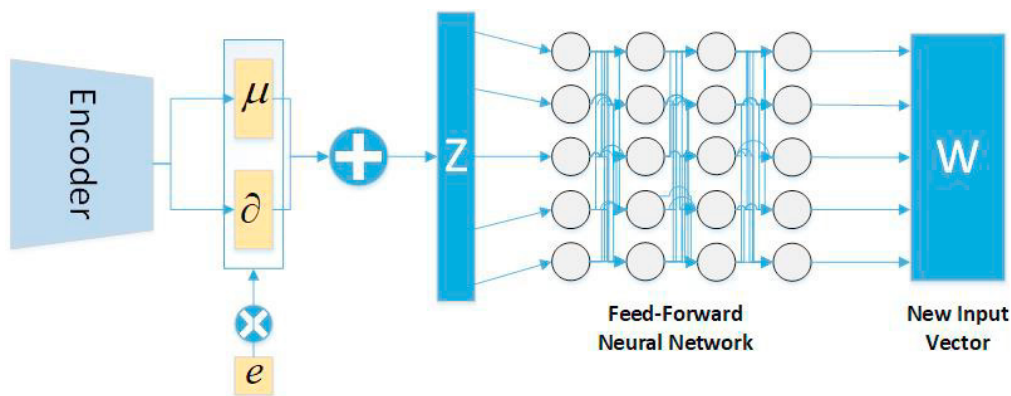


Fig. 4. Mapping Network of VAE-SGAN

3. Experimental results and analysis

3.1. Dataset

We chose the public database BCCD, which contains 12,500 enhanced images of blood cells, and the cell type label (CSV) contains 9,957 training images and 2,487 test images. We divided these cell images into four types:

eosinophils, lymphocytes, monocytes and neutrophils. In the original training image, there are 2,497 eosinophils, 2,483 lymphocytes, 2,478 monocytes, and 2,499 neutrophils [16]. Figures 5 shows images of different types of cells.

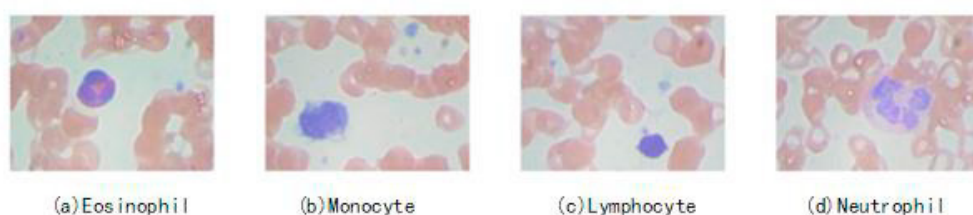


Fig. 5. Different kinds of white blood cell images

3.2. Lab environment

The experimental environment of this article is python3.6. Our processor is i9-9900k, memory is 32G, graphics card is GTX2080TI, and operating system is Linux. The whole experiment is based on the open source deep learning framework pytorch 0.4.

In the experiment, Adam optimizer is used for optimization. Among them, the batch size is $M=128$, and the input and output of all networks are scaled to $[-1,1]$. The learning rate of StyleGAN is set to 0.0002. The remaining parameter settings are designed according to the experimental parameters in DCGAN [17].

3.3. Result

The image generated by VAE is relatively blurry, and the image combined with StyleGAN is closer to the original image. Figure 6 shows the difference between the use of VAE and the combination of StyleGAN to generate pictures.

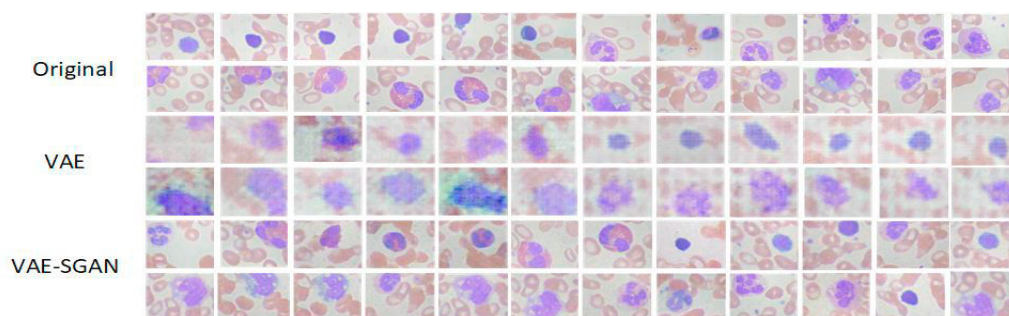


Fig. 6. VAE combined with StyleGAN to generate images

3.4. Qualitative analysis of generated images

In order to verify the experimental effect, this article compares the cell image generated by VAE and StyleGAN with the cell image generated by DCGAN and ProGAN methods. As shown in Figure 7, it can be found that the cell image generated by the VAE-SGAN method used in this article can firstly display the entire cell structure relatively clearly in structure, and secondly, in terms of specific details and textures, the rendering is more in line with the actual image. However, due to the lack of constraints, the other two methods do not have a clear image structure, the cell recognition is low, and the output smoothness is also poor.

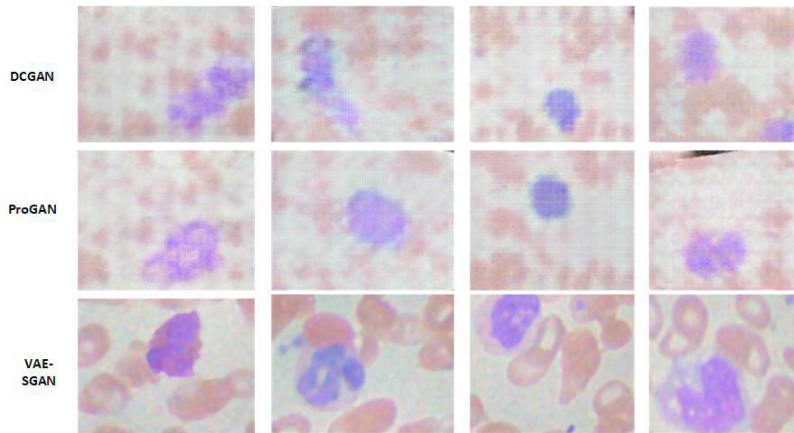


Fig. 7. Comparison of white blood cell images generated by different types of GANs

3.5. Generate image quantitative analysis

This paper uses the current international evaluation index IS (Inception Score) to evaluate the quality of the generated image. IS is one of the most commonly used methods to evaluate the image quality of GAN. The higher the value, the higher the quality of the generated image and the more realistic the generated image. It mainly examines the performance of two aspects of the image, one is whether the generated image is clear, and the other is whether the generated image is diverse [18].

In GAN, the expected conditional probability $P(y|x)$ can be highly predicted (x represents a given picture, y represents the main object contained in this picture), that is, the entropy value (used to describe the randomness of the transaction: if a random variable is highly predictable, then it has lower entropy; on the contrary, if it is highly unpredictable, then it uses higher entropy) lower. Then predict $P(y|x)$ that y is the label, so that it can represent the quality of the picture, the greater the probability, the better the quality. In terms of diversity, use the following formula to calculate the edge probability:

$$\int_z p(y|x=G(z))d_z \quad (2)$$

Which represents the output image generated by the generator with noise z . To sum up the two indicators, the calculation formula of IS is as follows:

$$IS(G) = e^{E_{x \sim p} D_{KL}(p(y|x)||p(y))} \quad (3)$$

Among them DKL is the calculation formula of KL-divergence.

In order to better verify the experimental effect of this paper, this paper uses DCGAN and ProGAN to generate 1000 cell image MNIST samples for 10 average evaluations. The experimental results are shown in Table 2-2.

Table 2. IS assessment results

Instructions	Real data	DCGAN	ProGAN	VAE-SGAN
IS	2.51	1.35	1.41	1.78

4. Conclusion

This article mainly elaborates the principle of Variational Autoencoder. Through this, the combination of VAE and StyleGAN is proposed. The network generates the hidden code of the image through VAE. As the input of StyleGAN, it can generate realistic cell images. Through quantitative and stereotyped experimental analysis, the results show that the proposed method is feasible. This method can generate high-quality cell images, thereby solving the problem of insufficient data. When there are enough high-quality medical data, artificial intelligence can find rules from big data through deep learning. However, the traditional diagnosis and treatment methods cannot do the best because of the limited ability of human beings. It can put forward meaningful suggestions to help doctors diagnose and reduce the rate of misdiagnosis.

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